



Universität Augsburg
Mathematisch-Naturwissenschaftlich-
Technische Fakultät

Classical networks for the Hubbard model with a tilted potential

- Master Thesis Colloquium -

Jonas Kell
Chair for theoretical Physics III

12th of March 2025

2025-03-12

Master Colloquium Presentation

1. Thank you for coming
2. Title: Classical networks for the Hubbard model with a tilted potential
3. Quickly what this will be about

Outline

1

Introduction of the Physical Problem

2025-03-12

Master Colloquium Presentation

└ Outline

1.
 - Introduction to the physical theory
 - In depth: Time evolution - why and how
 - Quickly: Sampling Strategy and simplifications
 - To give the title justice: Variational classical networks - what is all about
 - End (if there is time or if questions lead into that direction): Implementation and experiments
 - Outlook

Outline

1 Introduction of the Physical Problem



Outline

- 1 Introduction of the Physical Problem
- 2 Time Evolution



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- 2 Time Evolution
- 3 Sampling & Simplifications



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- 1 Introduction of the Physical Problem
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- 3 Sampling & Simplifications
- 4 Variational Classical Networks



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- 6 Conclusion & Outlook



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1	Introduction of the Physical Problem
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Particle Types

Introduction of the Physical Problem

■ Bosonic operators

$$\begin{aligned}\left[\hat{b}_l, \hat{b}_m\right] &= 0 \\ \left[\hat{b}_l^\dagger, \hat{b}_m^\dagger\right] &= 0 \\ \left[\hat{b}_l, \hat{b}_m^\dagger\right] &= \delta_{l,m}\end{aligned}$$



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└ Introduction of the Physical Problem

└ Particle Types

1. Wanna discuss, because during the writing of the thesis quite new insight in what in fact the differences of behavior are in between the particle types
2. First of all wanna point out how we have lattice operators, so fixed sites where there can be particles
3. Occupation number from 0 to infinity
4. Nice to handle because they commute

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- Fermionic operators

$$\{\hat{c}_l, \hat{c}_m\} = 0$$

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■ Commutation relations

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- Bosonic operators
 - Commutation relations
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- Hard-core bosonic operators
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 - Occupation number limited to 0 and 1
 - Number operator is idempotent
- Spins (Pauli operators/matrices)

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^2 = \hat{\sigma}^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}^3 = \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



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└ Introduction of the Physical Problem

└ Particle Types

1. Final particle type: spins - watch the 1/2 if it is a pauli or spin operator
2. Has also mostly commutative properties (also mainly *angular momentum* relations), but also one anti-commutation relation because of the properties of the Pauli matrices
3. Will be necessary for one mathematical re-formulation

Particle Types

Introduction of the Physical Problem

■ Bosonic operators

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■ Fermionic operators

■ Anti-commutation relations

■ Hard-core bosonic operators

■ Commutation relations

■ Occupation number limited to 0 and 1

■ Number operator is idempotent

■ Spins (Pauli operators/matrices)

$$\hat{\sigma}^x = 1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^y = \hat{\sigma}^z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

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Particle Types

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- Bosonic operators
 - Commutation relations
- Fermionic operators
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- Hard-core bosonic operators
 - Commutation relations
 - Occupation number limited to 0 and 1
 - Number operator is idempotent
- Spins (Pauli operators/matrices)
 - Particles from different sites commute
 - Different spin directions on the same site anti-commute
 - They obey angular momentum commutation relations

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\hat{\sigma}^3 = \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\left\{ \hat{\sigma}_l^\alpha, \hat{\sigma}_l^\beta \right\} = 2 \cdot \delta_{\alpha, \beta}$$

$$\left[\hat{\sigma}_l^\alpha, \hat{\sigma}_m^\beta \right] = 2 \cdot i \cdot \varepsilon_{\alpha, \beta, \gamma} \cdot \delta_{l, m} \cdot \hat{\sigma}_l^\gamma$$



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Particle Types

1. Final particle type: spins - watch the 1/2 if it is a pauli or spin operator
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Particle Types	
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■ Bosonic operators	$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
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■ Fermionic operators	$\hat{\sigma}^1 = \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
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■ Spins (Pauli operators/matrices)	
■ Particles from different sites commute	$\left\{ \hat{\sigma}_l^\alpha, \hat{\sigma}_l^\beta \right\} = 2 \cdot \delta_{\alpha, \beta}$
■ Different spin directions on the same site anti-commute	$\left[\hat{\sigma}_l^\alpha, \hat{\sigma}_l^\beta \right] = 2 \cdot i \cdot \varepsilon_{\alpha, \beta, \gamma} \cdot \delta_{l, m} \cdot \hat{\sigma}_l^\gamma$
■ They obey angular momentum commutation relations	

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices



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└ Introduction of the Physical Problem

└ Particle Types - Mappings

1. Pauli matrices are a base for the complex 2×2 matrices
2. Kronecker product of two pauli matrices therefore base for complex 4×4 matrices
3. Simple transformation that can be verified for its commutation relations or derived from two *Jordan-Wigner Transformations*
4. Important to watch the order of base states and keep convention or -1 will be introduced unexpectedly

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices
 - Base of the complex 2×2 matrices



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Particle Types - Mappings

Introduction of the Physical Problem

■ Properties of Pauli matrices

- Base of the complex 2x2 matrices
- Can be directly mapped to hard-core bosons

$$\hat{\sigma}_l^x = \hat{h}_l + \hat{h}_l^\dagger$$

$$\hat{\sigma}_l^y = i \cdot (\hat{h}_l - \hat{h}_l^\dagger)$$

$$\hat{\sigma}_l^z = 2 \cdot \hat{h}_l^\dagger \hat{h}_l - 1$$

$$|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |0\rangle$$

$$\hat{\sigma}^+ |\downarrow\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |1\rangle = \hat{h}^\dagger |0\rangle$$



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Hamiltonian of the system

Introduction of the Physical Problem

■ Hubbard model Hamiltonian

$$\mathcal{H} = \mathcal{H}_0 + \hat{V}$$

$$\mathcal{H}_0 = U \cdot \sum_l \hat{n}_{l,\uparrow} \hat{n}_{l,\downarrow} + \sum_{l,\sigma} \underbrace{\left(\vec{E} \cdot \vec{r}_l \right)}_{\varepsilon_l} \hat{n}_{l,\sigma}$$

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle, \sigma} \left(\hat{h}_{l,\sigma}^\dagger \hat{h}_{m,\sigma} + \hat{h}_{m,\sigma}^\dagger \hat{h}_{l,\sigma} \right)$$



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Master Colloquium Presentation

└ Introduction of the Physical Problem

└ Hamiltonian of the system

1. Hubbard model:

- Double occupation term
- Two particle degrees of freedom (= spin but not same spin as before: so also written as h and d)
- Here: electrical field that acts symmetrical on both spin directions
- Hopping term that is here treated as a perturbation

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0 + \hat{V} \\ \mathcal{H}_0 &= U \cdot \sum_l \hat{n}_l \hat{n}_l + \sum_{l,\sigma} \left(\vec{E} \cdot \vec{r}_l \right) \hat{n}_{l,\sigma} \\ \hat{V} &= -J \cdot \sum_{\langle l,m \rangle, \sigma} \left(\hat{h}_{l,\sigma}^\dagger \hat{h}_{m,\sigma} + \hat{h}_{m,\sigma}^\dagger \hat{h}_{l,\sigma} \right) \end{aligned}$$

Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field

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Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field
- Two particle types (spin degrees)

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 - Different *spins* than for Pauli spin operators

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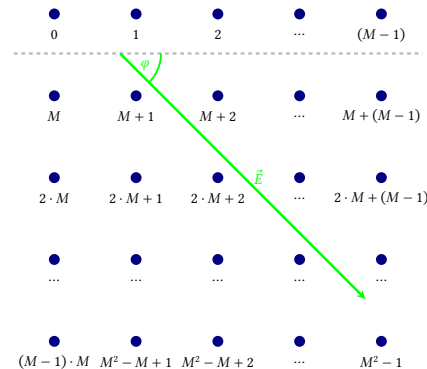
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- Two particle types (spin degrees)
 - Different spins than for Pauli spin operators

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0 + \hat{V} \\ \mathcal{H}_0 &= U \cdot \sum_l \hat{n}_l \hat{n}_l + \sum_{l,\sigma} \left(\vec{E} \cdot \vec{r}_l \right) \hat{n}_{l,\sigma} \\ \hat{V} &= -J \cdot \sum_{\langle l,m \rangle, \sigma} \left(\hat{h}_{l,\sigma}^\dagger \hat{h}_{m,\sigma} + \hat{h}_{m,\sigma}^\dagger \hat{h}_{l,\sigma} \right) \end{aligned}$$

Geometry of the system

Introduction of the Physical Problem

■ 2-dimensional geometry



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└ Introduction of the Physical Problem
└ Geometry of the system

Geometry of the system
Introduction of the Physical Problem

■ 2-dimensional geometry

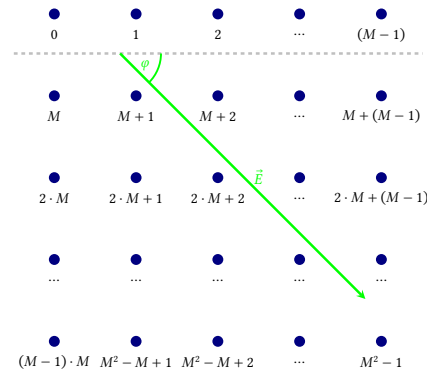


1. two-dimensional
2. square
3. Electrical field and we want to calculate generally for angle, field strength and number of sites

Geometry of the system

Introduction of the Physical Problem

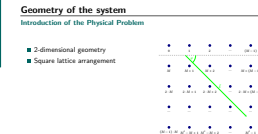
- 2-dimensional geometry
- Square lattice arrangement



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└ Introduction of the Physical Problem
└ Geometry of the system

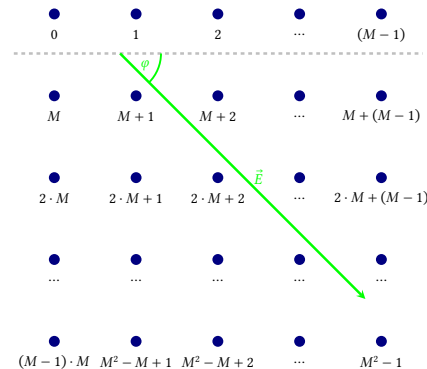
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Geometry of the system

Introduction of the Physical Problem

- 2-dimensional geometry
- Square lattice arrangement
- Computation for general size and field angle



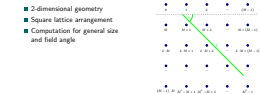
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- └ Introduction of the Physical Problem
- └ Geometry of the system

1. two-dimensional
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Geometry of the system
Introduction of the Physical Problem



Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables



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└ Time Evolution

└ Goal of the Calculation

1. Goal: evaluate the time-evolution for different observables (effectively)
2. Brute-force calculation is at least NP-hard, definitely exponential and one of the hardest problems to solve computationally generally
3. Strategy: perform a controlled expansion in the perturbation V and re-express the formulas as a dependency on a operator-free "effective Hamiltonian"
 - Can be re-ordered without concern as no operators
 - For the re-formulation we will use the interaction picture
4. Particle-type dependency is stored in this formula in the BASE STATES. Because in this re-formulation, the Heff is a pure number (no operator) and the initial state is a number (not operator dependent at all). But the base-state contains different info for fermions than for bosons, as the operators for hc-bosons commute and so the base-states are uniquely defined. But there are multiple possibilities for teh fermions, because swaps introduce minus signs. (State is not just 0s and 1s, but vacuum that is pre-set with operators and they have an order!!!!)

Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
 - Uses calculation in the *Interaction Picture*



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└ Time Evolution

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Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
- Uses calculation in the *Interaction Picture*

Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
 - Uses calculation in the *Interaction Picture*
 - Introduce operator-less *effective Hamiltonian*

$$|\Psi^S(t)\rangle = \sum_N e^{\mathcal{H}_{\text{eff}}(N,t)} \Psi_N |N\rangle$$



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Time Evolution

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Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
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$$|\Psi^S(t)\rangle = \sum_N e^{\mathcal{H}_{\text{eff}}(N,t)} \Psi_N |N\rangle$$

The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N, t) = -iE_0(N)t + \mathcal{H}_N(t)$$



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└ Time Evolution

└ The effective Hamiltonian

1. Comprised of two parts:

- Simple, quickly stated term that depends on the base energy
- Complicated one that will be expanded in the perturbation

The effective Hamiltonian

Time Evolution

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The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N, t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation



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The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N, t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation
- Base energy contribution:

$$\begin{aligned} E_0(N) &= \frac{\langle N | \mathcal{H}_0 | N \rangle}{\langle N | N \rangle} \\ &= \langle N | U \cdot \sum_l \hat{n}_{l,\uparrow} \hat{n}_{l,\downarrow} | N \rangle + \langle N | \sum_{l,\sigma} \varepsilon_l \hat{n}_{l,\sigma} | N \rangle \\ &= U \cdot \sum_l n_{l,\uparrow} n_{l,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} \end{aligned}$$



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Time Evolution

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The effective Hamiltonian

Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{h}_l^\dagger \hat{h}_m + \hat{h}_m^\dagger \hat{h}_l + \hat{d}_l^\dagger \hat{d}_m + \hat{d}_m^\dagger \hat{d}_l \right)$$



1. Need: V-Operator in the Interaction picture
2. For that: first solve the equations of motion for the ladder operators and the plug in
3. Solving the equations of motions in the interaction picture requires their number operator being *idempotent*

- To evaluate the time-evolution of the perturbation

$$\hat{V} = -J \cdot \sum_{\langle i, m \rangle} \left(\hat{h}_i^\dagger \hat{h}_m + \hat{h}_m^\dagger \hat{h}_i + \hat{d}_i^\dagger \hat{d}_m + \hat{d}_m^\dagger \hat{d}_i \right)$$

The effective Hamiltonian

Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{h}_l^\dagger \hat{h}_m + \hat{h}_m^\dagger \hat{h}_l + \hat{d}_l^\dagger \hat{d}_m + \hat{d}_m^\dagger \hat{d}_l \right)$$

- Solve the equation of motion for the ladder operators

$$\hat{h}_m^{\dagger I}(t) = e^{i \cdot \varepsilon_m \cdot t} \left(1 + (e^{iU \cdot t} - 1) \hat{d}_m^{\dagger S} \hat{d}_m^S \right) \hat{h}_m^{\dagger S}$$

$$\hat{h}_m^I(t) = e^{-i \cdot \varepsilon_m \cdot t} \left(1 + (e^{-iU \cdot t} - 1) \hat{d}_m^{\dagger S} \hat{d}_m^S \right) \hat{h}_m^S$$

$$\hat{d}_m^{\dagger I}(t) = e^{i \cdot \varepsilon_m \cdot t} \left(1 + (e^{iU \cdot t} - 1) \hat{h}_m^{\dagger S} \hat{h}_m^S \right) \hat{d}_m^{\dagger S}$$

$$\hat{d}_m^I(t) = e^{-i \cdot \varepsilon_m \cdot t} \left(1 + (e^{-iU \cdot t} - 1) \hat{h}_m^{\dagger S} \hat{h}_m^S \right) \hat{d}_m^S$$



- To evaluate the time-evolution of the perturbation

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{h}_l^\dagger \hat{h}_m + \hat{h}_m^\dagger \hat{h}_l + \hat{d}_l^\dagger \hat{d}_m + \hat{d}_m^\dagger \hat{d}_l \right)$$
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The effective Hamiltonian

Time Evolution

- Insert and reorder the operators

$$\begin{aligned}
 \hat{V}^I(t) &= \left\{ \hat{V}^S \right\}^I(t) = -J \cdot \sum_{[l,m]} \left\{ \left(\hat{h}_l^{\dagger S} \hat{h}_m^S + \hat{d}_l^{\dagger S} \hat{d}_m^S \right) \right\}^I(t) \\
 &= -J \cdot \sum_{[l,m]} \left(\hat{h}_l^{\dagger I}(t) \hat{h}_m^I(t) + \hat{d}_l^{\dagger I}(t) \hat{d}_m^I(t) \right) \\
 &= -J \cdot \sum_{[l,m]} \left[\Lambda_A(l, m, t) \cdot \hat{F}_A(l, m) + \Lambda_B(l, m, t) \cdot \hat{F}_B(l, m) + \Lambda_C(l, m, t) \cdot \hat{F}_C(l, m) \right]
 \end{aligned}$$



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Time Evolution

The effective Hamiltonian

1. Inserting yields TWO kinds of terms

- time-dependent analytical pre-factors (or their integral as one sees later)
- dressed hopping terms, basically only depending on the occupation of the state N

The effective Hamiltonian

Time Evolution

- Insert and reorder the operators

$$\begin{aligned}
 \hat{V}(t) &= \left\{ \hat{V}^S \right\}^I(t) = -J \cdot \sum_{[l,m]} \left\{ \left(\hat{h}_l^{\dagger S} \hat{h}_m^S + \hat{d}_l^{\dagger S} \hat{d}_m^S \right) \right\}^I(t) \\
 &= -J \cdot \sum_{[l,m]} \left(\hat{h}_l^{\dagger I}(t) \hat{h}_m^I(t) + \hat{d}_l^{\dagger I}(t) \hat{d}_m^I(t) \right) \\
 &= -J \cdot \sum_{[l,m]} \left[\Lambda_A(l, m, t) \cdot \hat{F}_A(l, m) + \Lambda_B(l, m, t) \cdot \hat{F}_B(l, m) + \Lambda_C(l, m, t) \cdot \hat{F}_C(l, m) \right]
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Time Evolution

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$$\Lambda_A(l, m, t) = e^{i(\varepsilon_l - \varepsilon_m) \cdot t} \quad \hat{F}_A(l, m) = \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l, \sigma}^{\dagger S} \hat{h}_{m, \sigma}^S \left(1 + 2 \cdot \hat{n}_{l, \bar{\sigma}}^S \hat{n}_{m, \bar{\sigma}}^S - \hat{n}_{l, \bar{\sigma}}^S - \hat{n}_{m, \bar{\sigma}}^S \right)$$

$$\Lambda_B(l, m, t) = e^{i(\varepsilon_l - \varepsilon_m + U) \cdot t} \quad \hat{F}_B(l, m) = \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l, \sigma}^{\dagger S} \hat{h}_{m, \sigma}^S \left(\hat{n}_{l, \bar{\sigma}}^S - \hat{n}_{l, \bar{\sigma}}^S \hat{n}_{m, \bar{\sigma}}^S \right)$$

$$\Lambda_C(l, m, t) = e^{i(\varepsilon_l - \varepsilon_m - U) \cdot t} \quad \hat{F}_C(l, m) = \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l, \sigma}^{\dagger S} \hat{h}_{m, \sigma}^S \left(\hat{n}_{m, \bar{\sigma}}^S - \hat{n}_{m, \bar{\sigma}}^S \hat{n}_{l, \bar{\sigma}}^S \right)$$



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Time Evolution

- # Master Colloquium Presentation

Time Evolution

- The effective Hamiltonian

1. Generally any expansion possible, but here we expand the *argument of the exponential* - cumulant
2. first order easily derivable with the info from the previous slide (integrate)
3. second and higher orders very complicated occupation terms and the pre-factors require quite a lot of edge-cases to integrate with the time-ordering operator
 - Hint onto the thesis for this exact derivation

The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian
 - Requires previously calculated value of the V-operator in the Interaction Picture

$$\begin{aligned}
 \mathcal{H}_N(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t dt_1 \int_0^t dt_2 \cdots \int_0^t dt_v \langle \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) \cdots \hat{V}^I(t_v) \rangle_{c(N)} \\
 &= -i \int_0^t dt_1 \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} \\
 &\quad - \frac{1}{2} \int_0^t dt_1 \int_0^t dt_2 \left(\frac{\langle N | \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} - \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle \cdot \langle N | \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle^2} \right) \\
 &\quad + \cdots
 \end{aligned}$$



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Master Colloquium Presentation

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 &\approx -i \int_0^t dt_1 \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} \\
 &\quad - \frac{1}{2} \int_0^t dt_1 \int_0^t dt_2 \left(\frac{\langle N | \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} - \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle \cdot \langle N | \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle^2} \right) \\
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The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian
 - Requires previously calculated value of the V-operator in the Interaction Picture
 - Controllable *cumulant expansion*

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 &= -i \int_0^t dt_1 \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} \\
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- Evaluate the contribution to the effective Hamiltonian

- Requires previously calculated value of the V-operator in the Interaction Picture
- Controllable cumulant expansion

$$\begin{aligned}
 H_E(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t dt_1 \int_0^t dt_2 \cdots \int_0^t dt_v \langle \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) \cdots \hat{V}^I(t_v) \rangle_{c(N)} \\
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 \end{aligned}$$

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} =$$

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Time Evolution

Handling of Observables

1. Why? General evaluation of an expectation value for an observable.
 - Probability for state
 - Local observable (can be swapped for many different ones)
2. Does ONLY require difference of effective Hamiltonian for evaluation (notice for later)

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} =$$

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values
 - Requires a probability for the state
 - Requires a local observable

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} = \sum_N P(N, t) \underbrace{\sum_K \langle N | \hat{\mathcal{O}} | K \rangle e^{\mathcal{H}_{\text{eff}}(K, t) - \mathcal{H}_{\text{eff}}(N, t)} \frac{\Psi_K}{\Psi_N}}_{\hat{\mathcal{O}}_{\text{loc}}(N, t)} = \sum_N P(N, t) \hat{\mathcal{O}}_{\text{loc}}(N, t)$$



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Time Evolution

Handling of Observables

1. Why? General evaluation of an expectation value for an observable.
 - Probability for state
 - Local observable (can be swapped for many different ones)
2. Does ONLY require difference of effective Hamiltonian for evaluation (notice for later)

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values
 - Requires a probability for the state
 - Requires a local observable

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} = \sum_N P(N, t) \underbrace{\sum_K \langle N | \hat{\mathcal{O}} | K \rangle e^{\mathcal{H}_{\text{eff}}(K, t) - \mathcal{H}_{\text{eff}}(N, t)} \frac{\Psi_K}{\Psi_N}}_{\hat{\mathcal{O}}_{\text{loc}}(N, t)} = \sum_N P(N, t) \hat{\mathcal{O}}_{\text{loc}}(N, t)$$

Simple Observables

Time Evolution

- Local observable for double occupation measurement

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└ Time Evolution

└ Simple Observables

1. Complex valued observable, make sure this cancels in the case of approximations
2. *SHOULD MENTION*: Energy and variance (most expensive ones, are assumed to keep constant, can be used to gauge the quality of the expansion)

Simple Observables

Time Evolution

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Simple Observables

Time Evolution

- Local observable for double occupation measurement
 - Observables generally are very sparse matrices
 - Reduces to pure occupation measurement

$$\hat{\mathcal{O}}_{\text{loc}}(N, t) = \sum_K \langle N | \hat{\mathcal{O}}_{\text{do}}(l) | K \rangle e^{\mathcal{H}_{\text{eff}}(K, t) - \mathcal{H}_{\text{eff}}(N, t)} \frac{\Psi_K}{\Psi_N} = n_{l, \uparrow} \cdot n_{l, \downarrow}$$



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 - Requires evaluation of the difference of two effective Hamiltonians

$$\hat{\mathcal{O}}_{\text{loc}}(N, t) = i \cdot \left[n_{m, \sigma} \cdot (1 - n_{l, \sigma}) - n_{l, \sigma} \cdot (1 - n_{m, \sigma}) \right] \cdot \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N}, t) - \mathcal{H}_{\text{eff}}(N, t)}$$



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Access to Density-Matrices

Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix

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└ Time Evolution

└ Access to Density-Matrices

1. Quite interesting, because of this extra mentioned the Pauli operators
2. No access to the full density matrix, can not trace out, because too many states to trace out
3. BUT: as the complex 4x4 matrix can be written in the base of the kronecker product of two pauli matrices, can be expressed (because the pauli operators can be translated into hard-core bosonic ones again and for these we have the general formula for expectation values)



Access to Density-Matrices

Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
 - Access to *purity*, *concurrence* and other entanglement measures/monotones



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└ Time Evolution

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Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
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- Direct calculation not possible

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_{k=0}^M (\mathbb{1}_A \otimes \langle \chi_k |) \rho_{AB} (\mathbb{1}_A \otimes | \chi_k \rangle)$$



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Time Evolution

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- Use Pauli matrices to expand the complex 4x4 matrix

$$\rho_A(t) = \rho_{l,\sigma,m,\mu}(t) = \frac{1}{4} \sum_{\alpha,\beta \in \{0,x,y,z\}} \langle \hat{\sigma}_{l,\sigma}^\alpha \hat{\sigma}_{m,\mu}^\beta \rangle_{(t)} (\hat{\sigma}^\alpha \otimes \hat{\sigma}^\beta)$$



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Access to Density-Matrices

Time Evolution

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 \hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^x \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) &= \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \\
 \hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^y \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) &= \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma}) \\
 \hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^z \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) &= \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot (2 \cdot n_{l,\sigma} - 1) \\
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 \end{aligned}$$



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Time Evolution

Access to Density-Matrices

1. Sixteen terms (with missing cases trivially derived from the ones that are given)
2. Again all depend on the difference of effective hamiltonians between two states ith local modifications

Access to Density-Matrices

Time Evolution

$$\begin{aligned}
 \hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^x \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) &= \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \\
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 \end{aligned}$$

Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization



1. Normalization requires knowing exponential amount of states and we need to sample an exponential amount to get all
2. switch to METROPOLIS HASTINGS algorithm
 - only fixed number of samples required (depends on the desired precision)
 - No normalization needed, because it cancels in the division

Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization
- Switch to the Metropolis *Hastings* algorithm
 - Non-exponential number of samples

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} = \sum_N P(N, t) \cdot \hat{\mathcal{O}}_{\text{loc}}(N, t) \approx \frac{1}{|\{N\}_{\text{MC}}|} \sum_{\{N\}_{\text{MC}}} \hat{\mathcal{O}}_{\text{loc}}(N, t)$$



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Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization
- Switch to the Metropolis *Hastings* algorithm
 - Non-exponential number of samples
 - Depends only on a transition probability
 - Does not need normalization because it cancels

$$\begin{aligned} \alpha &= \frac{P(\tilde{N}, t)}{P(N, t)} = \frac{f(\tilde{N}, t)}{f(N, t)} = \frac{|e^{\mathcal{H}_{\text{eff}}(\tilde{N}, t)}|^2 |\Psi_{\tilde{N}}|^2}{|e^{\mathcal{H}_{\text{eff}}(N, t)}|^2 |\Psi_N|^2} \\ &= \frac{|\Psi_{\tilde{N}}|^2 e^{\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) + i\Im(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) + \Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) - i\Im(\mathcal{H}_{\text{eff}}(\tilde{N}, t))}}{|\Psi_N|^2 e^{\Re(\mathcal{H}_{\text{eff}}(N, t)) + i\Im(\mathcal{H}_{\text{eff}}(N, t)) + \Re(\mathcal{H}_{\text{eff}}(N, t)) - i\Im(\mathcal{H}_{\text{eff}}(N, t))}} \\ &= \frac{|\Psi_{\tilde{N}}|^2}{|\Psi_N|^2} e^{2\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) - 2\Re(\mathcal{H}_{\text{eff}}(N, t))} \\ &= \frac{|\Psi_{\tilde{N}}|^2}{|\Psi_N|^2} e^{2\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t) - \mathcal{H}_{\text{eff}}(N, t))} \end{aligned}$$

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Analytical Simplifications

Sampling & Simplifications

- Choose a helpful initial state

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└ Sampling & Simplifications

└ Analytical Simplifications

Analytical Simplifications

Sampling & Simplifications

■ Choose a helpful initial state

1. Choose the helpful *initial state* with all base states having the same probability
2. because of this the divisions of Ψ_N over Ψ_K everywhere give 1 and many terms cancel (outside of the local interaction range of the modification)



Analytical Simplifications

Sampling & Simplifications

- Choose a helpful initial state

$$\Psi_N = \frac{1}{\sqrt{\#(\text{states})}} = \frac{1}{\sqrt{2^{\#(\text{sites})} \cdot 2}} = \frac{1}{2^{\#(\text{sites})}}$$

$$|\Psi^S(t=0)\rangle = \bigotimes_{l=1}^{\#(\text{sites})} \frac{1}{2} \left(1 + \hat{h}_{l,\uparrow}^{\dagger S} + \hat{h}_{l,\downarrow}^{\dagger S} + \hat{h}_{l,\uparrow}^{\dagger S} \hat{h}_{l,\downarrow}^{\dagger S} \right) |0\rangle$$

- Now *almost all* terms cancel in *differences* of the effective Hamiltonian



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└ Sampling & Simplifications

└ Analytical Simplifications

Analytical Simplifications

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- Now *almost all* terms cancel in *differences* of the effective Hamiltonian

- Exemplary for single flip on base energy term

$$\begin{aligned} E_0(N) - E_0(\tilde{N}) &= U \sum_l n_{l,\downarrow} n_{l,\uparrow} - U \sum_l \tilde{n}_{l,\downarrow} \tilde{n}_{l,\uparrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} - \sum_{l,\sigma} \varepsilon_l \tilde{n}_{l,\sigma} \\ &= \varepsilon_i (2n_{i,\sigma_i} - 1) + U \cdot \begin{cases} \sigma_i = \uparrow : & n_{i,\downarrow} (2n_{i,\uparrow} - 1) \\ \sigma_i = \downarrow : & n_{i,\uparrow} (2n_{i,\downarrow} - 1) \end{cases} \end{aligned}$$



- Choose a helpful initial state

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- Now almost all terms cancel in differences of the effective Hamiltonian

$$\begin{aligned} E_0(N) - E_0(\tilde{N}) &\approx U \sum_l n_{l,\downarrow} n_{l,\uparrow} - U \sum_l \tilde{n}_{l,\downarrow} \tilde{n}_{l,\uparrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} - \sum_{l,\sigma} \varepsilon_l \tilde{n}_{l,\sigma} \\ &= \varepsilon_i (2n_{i,\sigma_i} - 1) + U \cdot \begin{cases} \sigma_i = \uparrow : & n_{i,\downarrow} (2n_{i,\uparrow} - 1) \\ \sigma_i = \downarrow : & n_{i,\uparrow} (2n_{i,\downarrow} - 1) \end{cases} \end{aligned}$$

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Why and How should Parameters be variational?

Variational Classical Networks

- Theoretically: always better approximation through higher order terms



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└ Variational Classical Networks

└ Why and How should Parameters be variational?

Why and How should Parameters be variational?

Variational Classical Networks

- Theoretically: always better approximation through higher order terms

1. Argument:

- Could go on performing the controlled expansion for all perturbation-strengths and get nice result
- But higher orders complicated and expensive
- Try to get more from the lower orders, by using Variational parameters (optimized by the problem, we hope they can incorporate influence from higher order terms without them being present)

2. Time dependent variational principle (TDVP)

- Find a parametrized *wavefunction*
- Update the parameters so that a tiny nudge into the direction of the *time DERIVATIVE* of the parameter vector has the same effect as a time evolution with the full hamiltonian
- The first order numerical integration (could do higher orders, but for the concept here)

3. O: Variational derivative E: local energy (we had already earlier)

Why and How should Parameters be variational?

Variational Classical Networks

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└ Variational Classical Networks

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Variational Classical Networks

- Theoretically: always better approximation through higher order terms
 - Hard to calculate analytically and expensive to evaluate
- Hope: Better behavior if parameters are *learned/optimized*, starting from the analytical result



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└ Variational Classical Networks

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3. O: Variational derivative E: local energy (we had already earlier)

Why and How should Parameters be variational?

Variational Classical Networks

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2. S: covariance matrix
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First Try: Cumulant Expansion Prefactors

Variational Classical Networks

- Idea: generate a variational Hamiltonian by replacing the analytical prefactors

$$\mathcal{H}_N(t) = \sum_i C_i(t) \cdot \Phi_i(N) \quad \leftrightarrow \quad \mathcal{H}_{\text{VCN}}(N, \vec{\eta}) = \sum_i \eta_i \cdot \Phi_i(N)$$

1. Idea: replace prefactors that depend on the time and keep the shape of the terms that depend on the occupation of the state
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Variational Classical Networks

- First strategy is not suitable
 - Energy- and variance is not conserved

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1. Fails because not taken the explicit time-dependency into account as it was *differentiated* away wrongly
2. As a solution: introduce variational base-energy factors - which is a successful strategy

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Implementation & Experiments

- Too much code to properly show
 - Look at repository for overview



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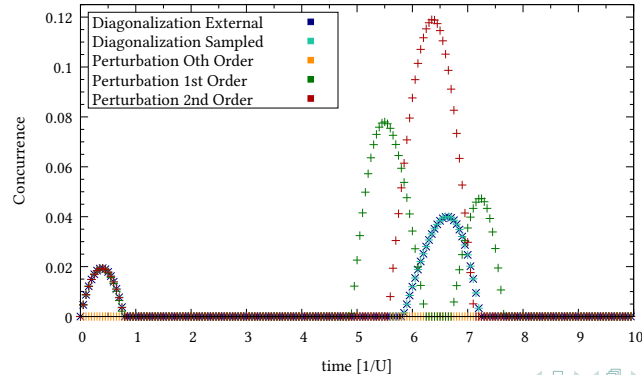
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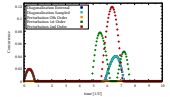
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- Working implementation
 - Extensively tested
 - Configurable and extensible
- Started validating expectations with measurements on the compute cluster

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2. All tools and other resources are referenced in the presentation
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References I

Summary & Conclusion

- [1] J. Kell, *Latex sources for this presentation and the main thesis*, (2025)
<https://github.com/jonas-kell/master-thesis-documents>.
- [2] J. Kell, *Thesis code implementation and reference*, (2025)
<https://github.com/jonas-kell/master-thesis-code>.
- [3] J. Kell, *Https://github.com/jonas-kell/beamer-theme-unia-mntf*, (2024)
<https://github.com/jonas-kell/math-manipulator>.



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└ Summary & Conclusion

└ References

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- [3] J. Kell, *Https://github.com/jonas-kell/beamer-theme-unia-mntf*, (2024)
<https://github.com/jonas-kell/math-manipulator>.

Extra Plots: Overview

Extra slides

- Exact comparison density matrix
- First orders: energy and variance comparison
- J parameter sweep
- Monte Carlo sampling convergence
- Runtime validation
- VCN: parametrization failiure
- VCN: energy conservation can be controlled
- VCN: minimal square reference system
- VCN: larger square reference system in direct comparison
- VCN: system size dependency



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└ Extra slides

└ Extra Plots: Overview

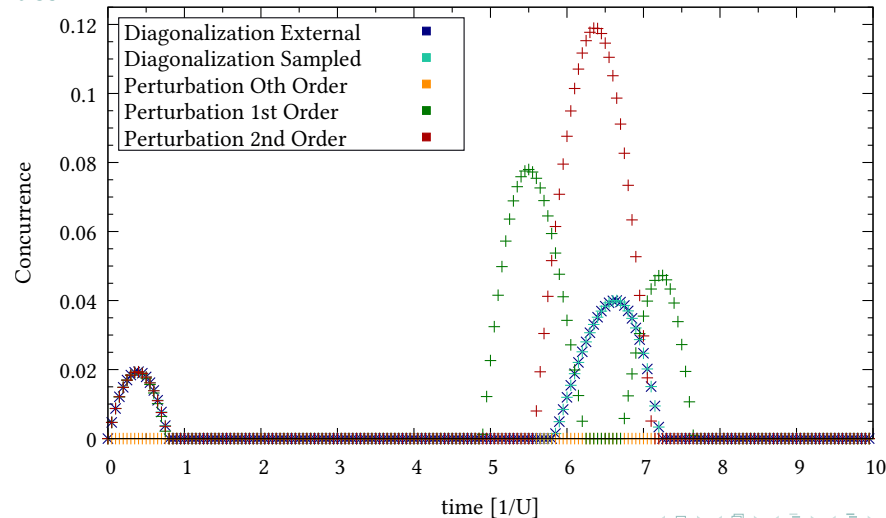
Extra Plots: Overview

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Exact Comparison: Concurrence

Extra slides

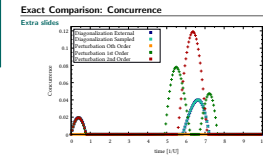


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Extra slides

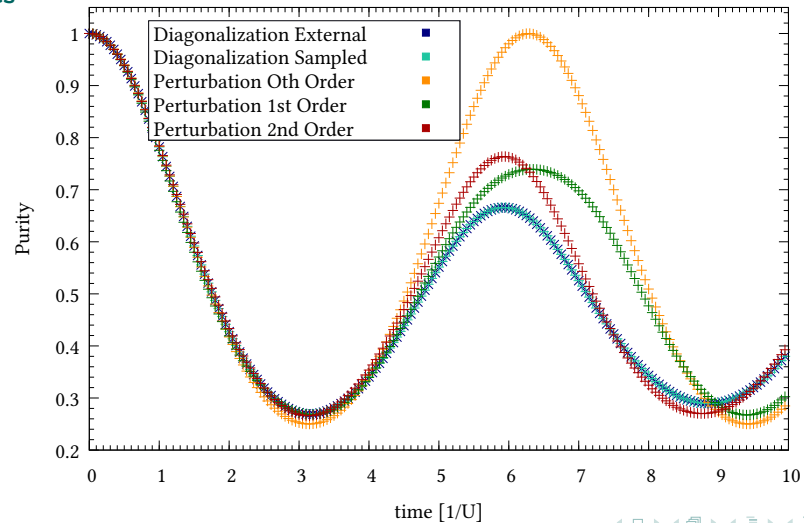
Exact Comparison: Concurrence



1. Concurrence comparison of small system with exact calculations and comparison thereof
2. See extreme boost of second order
3. NOTICE: how the External calculation is done COMPLETELY on density matrices and diagonalization, so proves we can sample the reduced density matrix out

Exact Comparison: Purity

Extra slides



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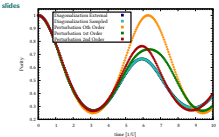
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└ Extra slides

└ Exact Comparison: Purity

Exact Comparison: Purity

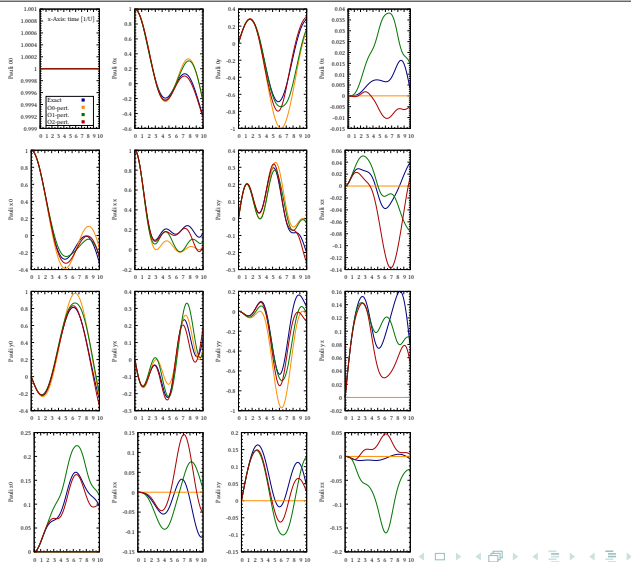
Extra slides



1. Purity measurements for the graph of the measurement from one page before

Exact Comparison: Pauli Measurements

Extra slides



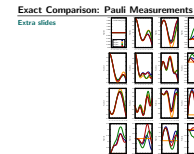
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└ Extra slides

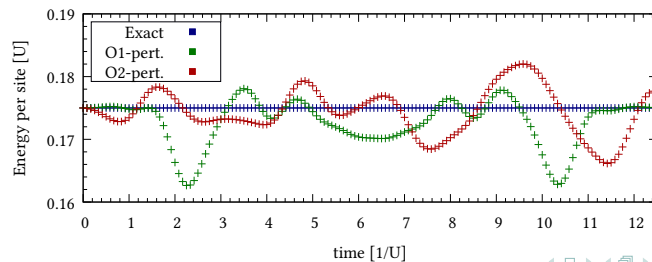
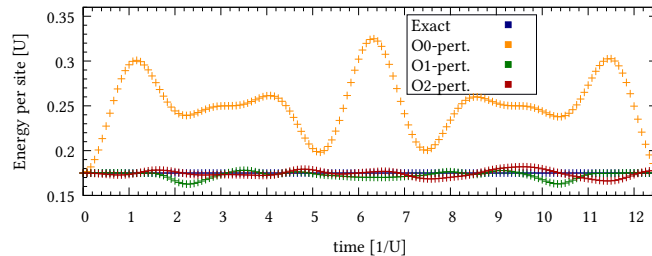
└ Exact Comparison: Pauli Measurements

1. Portrait mode format, because extracted from the thesis.
2. You can see what a problem the z-contributions are



First Orders: Energy Comparison

Extra slides



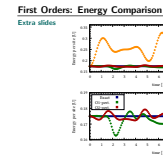
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└ Extra slides

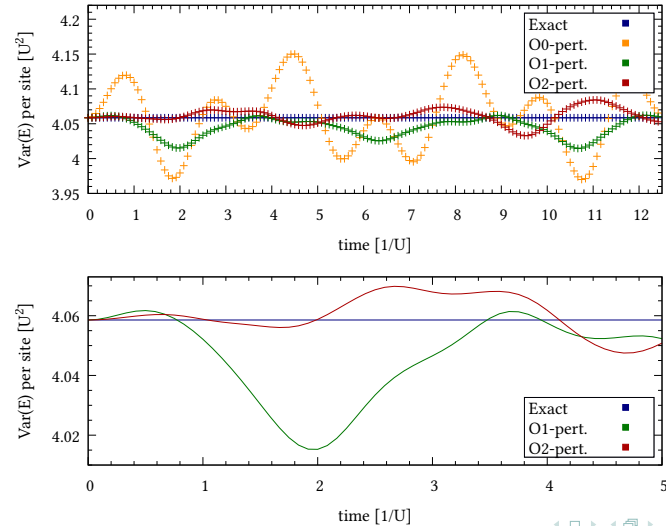
└ First Orders: Energy Comparison

1. Second picture zoomed in



First Orders: Variance Comparison

Extra slides



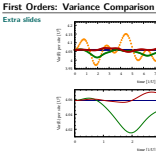
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└ Extra slides

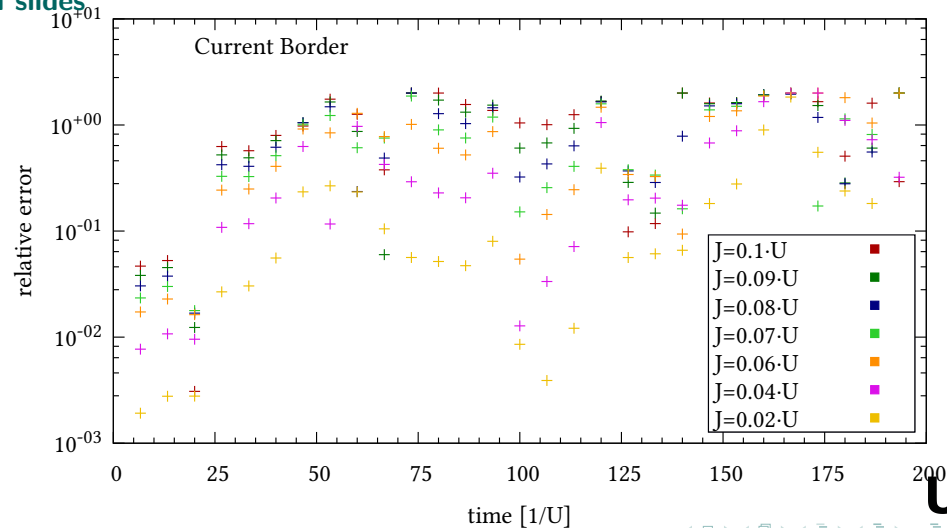
└ First Orders: Variance Comparison

1. Second picture zoomed in



J-Sweep: Current Border

Extra slides



UNIA

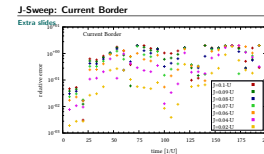
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Master Colloquium Presentation

Extra slides

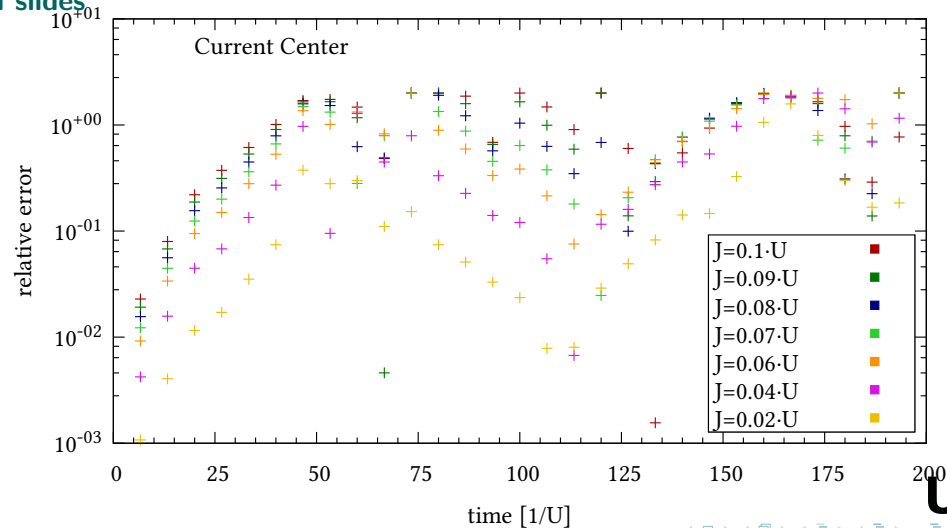
J-Sweep: Current Border

1. Logarithmic plot
2. Measurements show the trend of the convergence of the relative error for small perturbations
3. Depends on the observable though, still



J-Sweep: Current Center

Extra slides

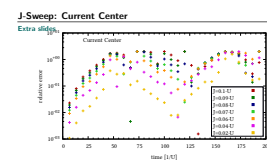


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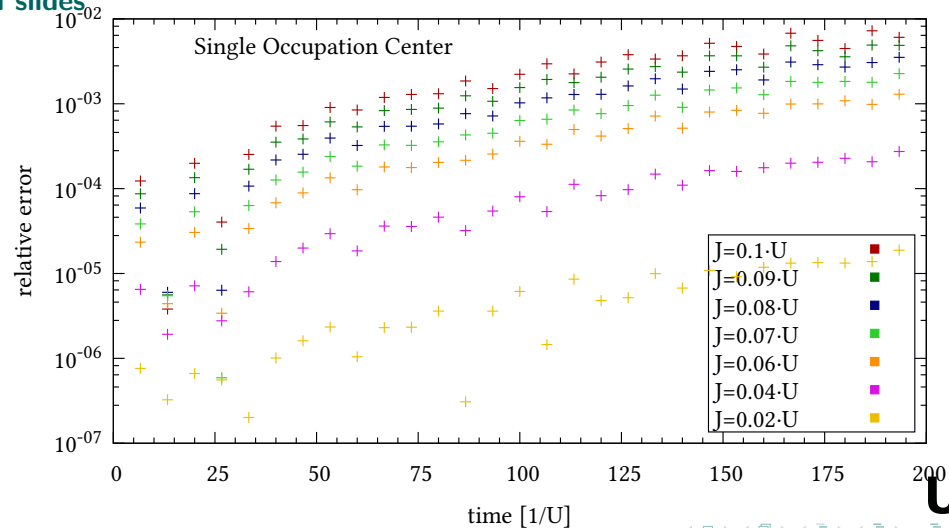
Extra slides

J-Sweep: Current Center



J-Sweep: Single Occupation Center

Extra slides



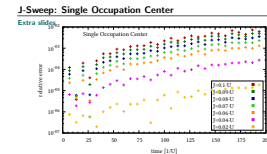
UNIA

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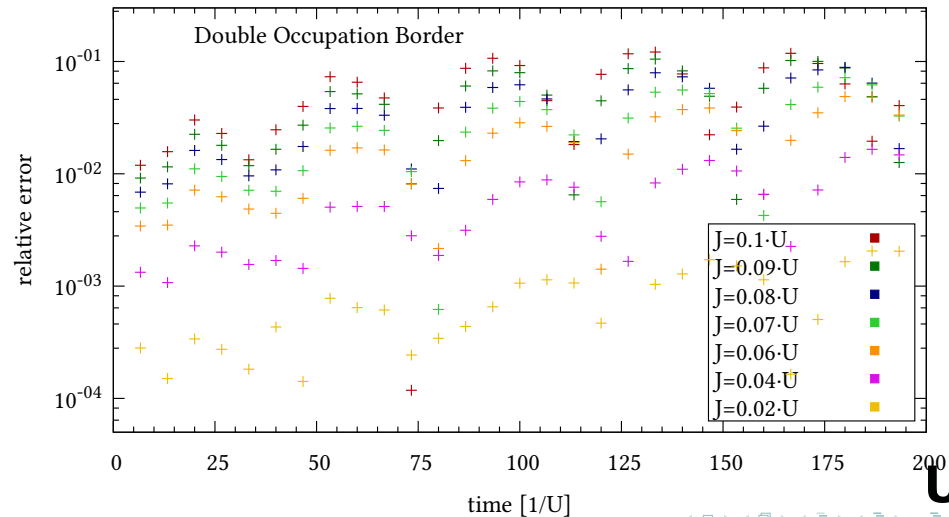
Extra slides

J-Sweep: Single Occupation Center



J-Sweep: Double Occupation Border

Extra slides

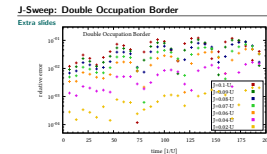


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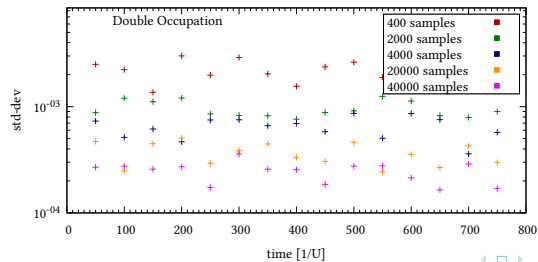
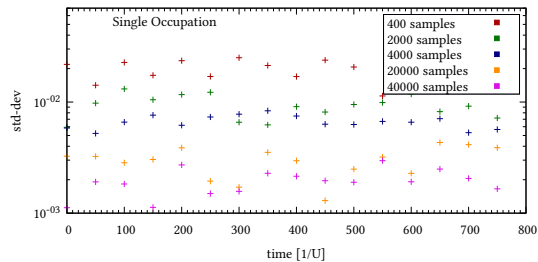
Extra slides

J-Sweep: Double Occupation Border



Monte Carlo Convergence: Occupation

Extra slides



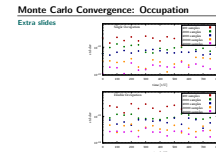
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Master Colloquium Presentation

Extra slides

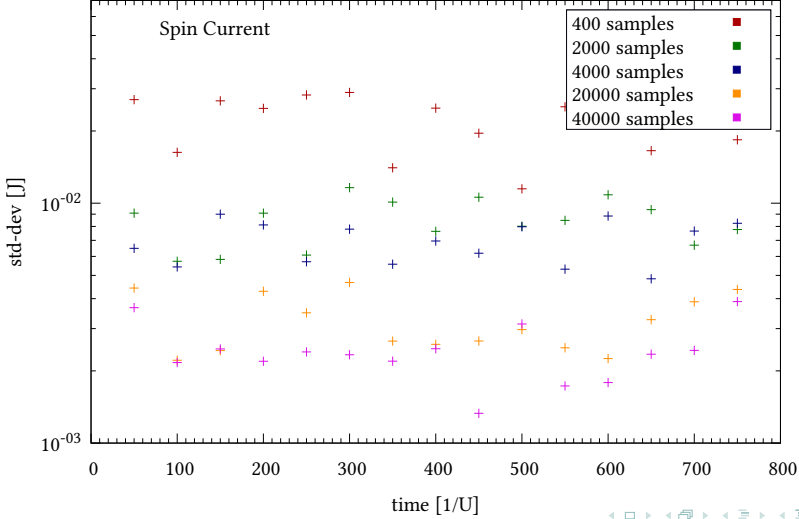
Monte Carlo Convergence: Occupation

1. Well as we can see, the appriximation converges as the variance gos to zero



Monte Carlo Convergence: Occupation

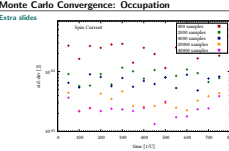
Extra slides



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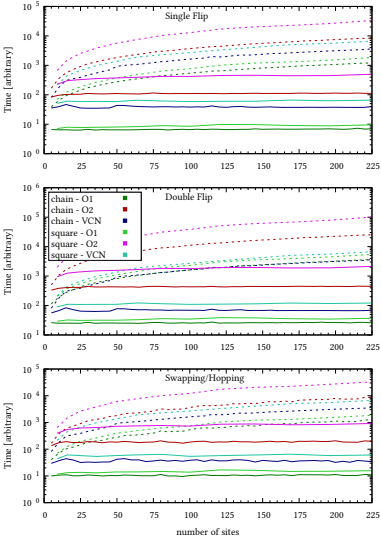
Master Colloquium Presentation

- Extra slides
- Monte Carlo Convergence: Occupation



Runtime Validation

Extra slides



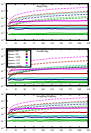
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└ Extra slides

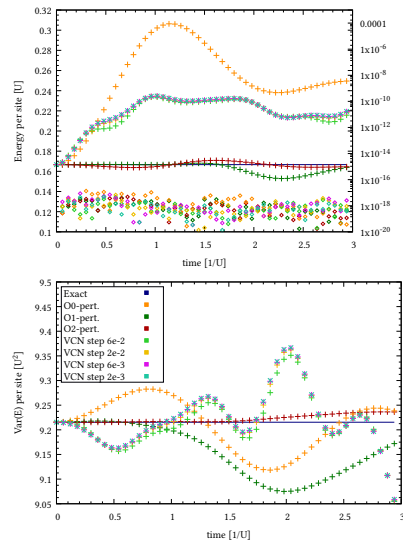
└ Runtime Validation

Runtime Validation
Extra slides



VCN Failure: Non-Explicit Parametrization

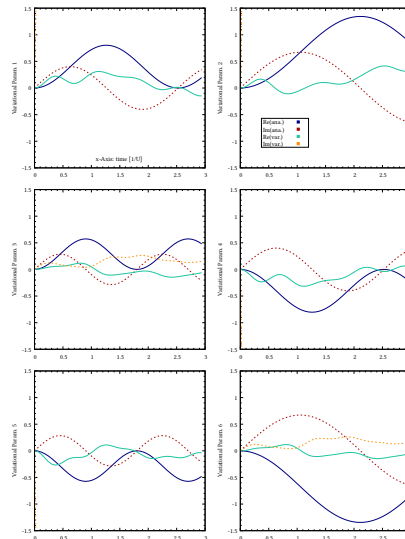
Extra slides



1. Complex part now properly vanishes (was long a problem)
2. See how the vcn goes like the base energy curve only in the beginning (suspiciously)

VCN Failure: Non-Explicit Parametrization

Extra slides



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Master Colloquium Presentation

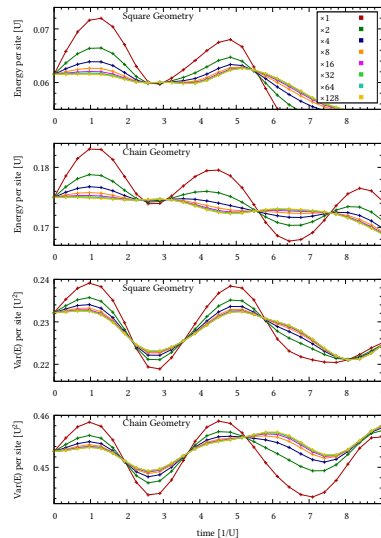
└ Extra slides

└ VCN Failure: Non-Explicit Parametrization

1. See complex part jetting off to infinity because it doesn't know how to compensate

VCN Validation: Energy Conservation Controllable

Extra slides



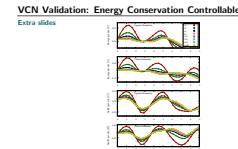
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└ Extra slides

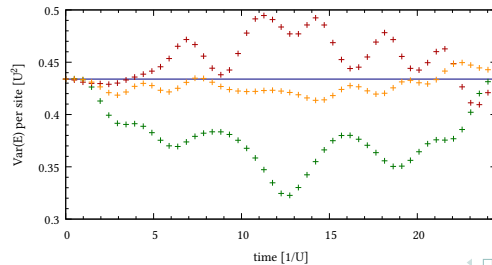
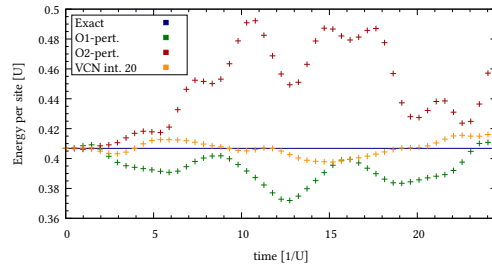
└ VCN Validation: Energy Conservation Controllable

1. Still no perfect conservation for infinite steps, but this is probably due to first order integration only

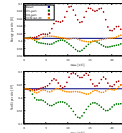


VCN: Small Square System

Extra slides

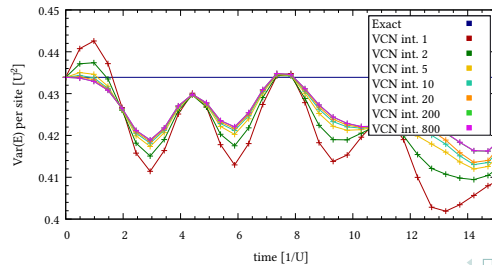
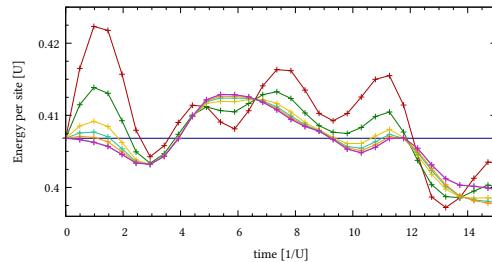


1. Notice how VCN now is MUCH better

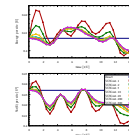


VCN: Small Square System

Extra slides

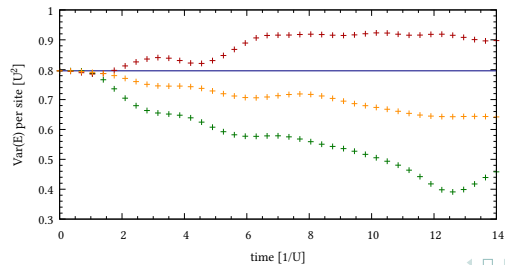
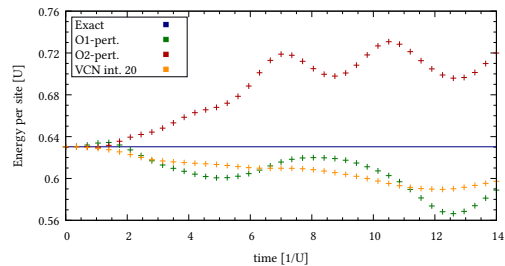


1. Step size dependent version of previous slide



VCN: Bigger Square System

Extra slides



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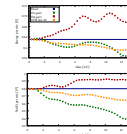
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└ Extra slides

└ VCN: Bigger Square System

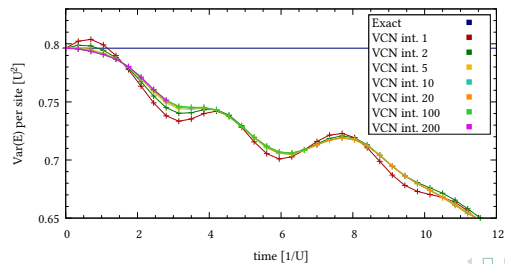
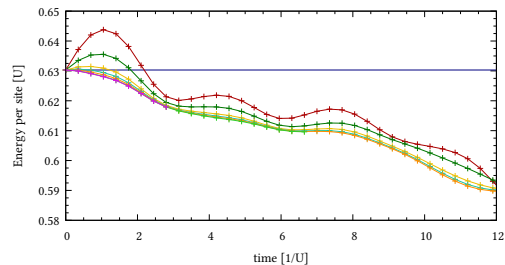
1. Now not a guarantee for immediately better results, but very promising

VCN: Bigger Square System
Extra slides



VCN: Bigger Square System

Extra slides



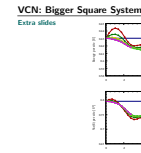
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└ Extra slides

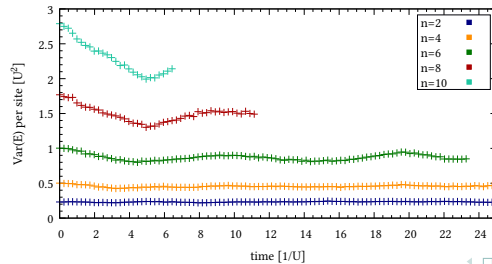
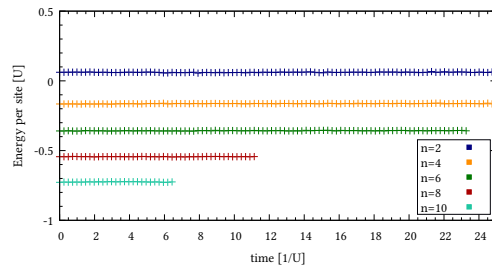
└ VCN: Bigger Square System

1. Here one can see very quickly more steps doesn't only make it better



VCN End to End Test: System Size Dependency

Extra slides



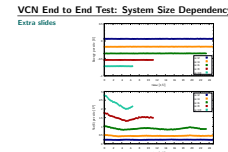
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Master Colloquium Presentation

└ Extra slides

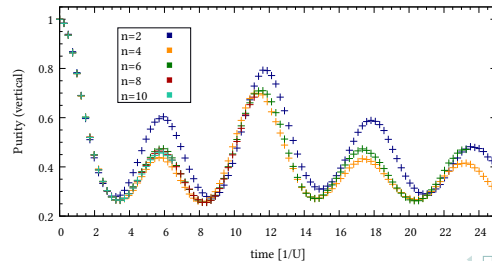
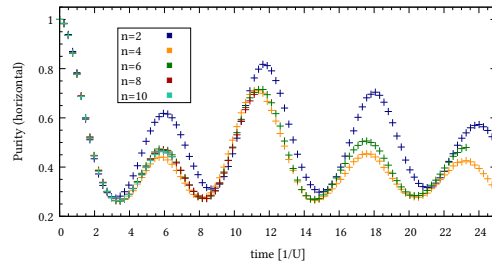
└ VCN End to End Test: System Size Dependency

1. This is the energy for validation



VCN End to End Test: System Size Dependency

Extra slides



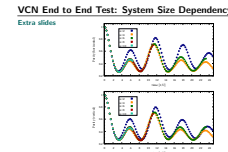
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└ Extra slides

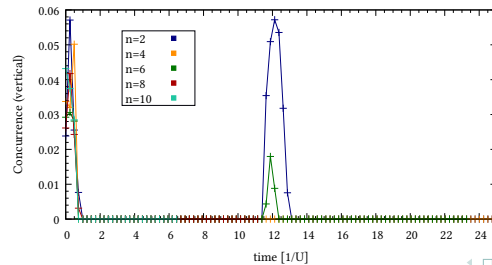
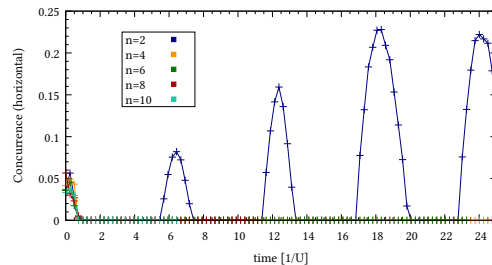
└ VCN End to End Test: System Size Dependency

1. This is the purity
2. Does not seem to have a huge variation



VCN End to End Test: System Size Dependency

Extra slides



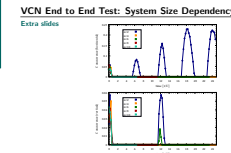
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Extra slides

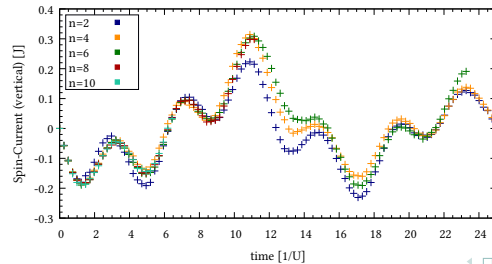
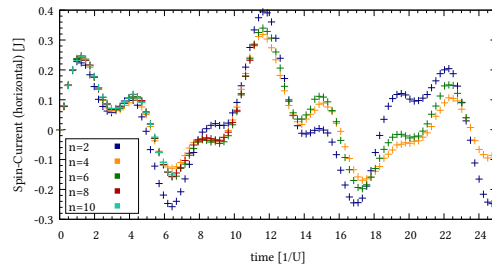
VCN End to End Test: System Size Dependency

1. This is the concurrence
2. Poorly resolved, this should be looked into possibly in the future



VCN End to End Test: System Size Dependency

Extra slides



1. This is the current
2. Most obvious effect, but still extremely stable over a order of magnitude of sites
3. See assymmetric behavior, because electrical field is set to not be symmetric